

A Historical Note

It is plausible that precisely because inversion of the plane in a circle is not a very difficult notion that requires only a minimal amount of basic mathematical knowledge, it was invented/discovered across time by multiple people at multiple locales on our planet.

For example. In Western Europe, as hard physical evidence suggests, in the proposition **I.37** of his work “*Conics*” the ancient Greek mathematician Apollonius of Perga (240 BC - 190 BC) defines not only the inversion of the plane with respect to a circle, but he also defines the inversion of the plane in an ellipse and in a hyperbola.

Technically, in his “*Conics*”, Apollonius did not explicitly enumerate the properties of inversion of the real plane in a circle but, judging by how another ancient Greek mathematician known as Pappus of Alexandria (290 AD - 350 AD) retold the relevant statements that Apollonius did put forward explicitly, it is evident that Apollonius simply had to understand and know well that under the inversion of the plane with respect to a circle the preimage circles are taken into circles.

In Western Europe still, apparently and according to Coxeter, “*Introduction to Geometry*”, second edition, p. 77, a German Jewish mathematician L. I. Magnus (1790-1861) invented inversion of the plane in a circle in 1831.

On the other side of the globe, as we show in the Problem 26, Inversion Meets Sangaku, it is evident that the Japanese mathematicians of the past had to be aware of inversion of the plane in a circle as well.